

Dissertation submission note

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For Dr. Songa Mutambi, 21 September 2026

Measuring the External Field: A Cortical-Proximal Content Observable and the Mean-Field Bound on Media-Driven Opinion Change. 81 pages, seven chapters, 21 references. The title and scope follow the amendment you approved.

Changes since your last review

1. **Baselines (Chapter 5).** On the same 400 items and label, TF-IDF with logistic regression reaches AUC 0.9758 against the index's 0.6274; VADER sentiment does not clear chance. The thesis states that the index is a field observable, not a competitive classifier.
2. **Leave-one-source-out (Chapter 5).** Not computable: every source dataset carries exactly one class. The 150 ISOT items, the one source with both classes, are used instead; TF-IDF still reaches 0.9724 there.
3. **The outrage result (Chapters 5 and 6).** High outrage raises both networks almost equally ($d = +0.507$ and $+0.491$), so its flat index is a symmetric response, not an absent one. The earlier argument that this argues against pure source detection is withdrawn.
4. **Sign reversals (Chapter 6).** 51 of 400 items reverse between sessions; the measured session noise predicts 55.3, 95% interval [44, 67]. No per-item claim is made.
5. **Citations and wording.** Every dataset, model and atlas is now cited where it is used. The bound $\alpha \geq h_c(\beta J) / \Delta X$ is stated as a result for the all-to-all mean-field model, since network structure lowers h_c .

Results as submitted

Category separation	$\eta^2 = 0.1068$, $p = 1.035 \times 10^{-9}$ (detectable 0.0268)
Second session	ICC 0.8725, separation replicates at $\eta^2 = 0.0888$
Main confound	each category comes from one source dataset, stated before every result
Coupling α	unidentified; no value quoted
Field bound	$\Delta X = 0.1241$ gives $\alpha \geq 4.2938$ at $\beta J = 2$

Stated as limitations, not claimed

Whether the encoder's predictions track measured cortex (the noise ceiling, 0.1517, is measured; the prediction pass against it is not yet run). A corpus that crosses sources with categories. Any subcortical structure, which the released checkpoint does not predict.

What I am asking for

1. Any corrections you require before the thesis is final.
2. Your signature on the supervisor's declaration (page iii).
3. Clearance to deposit the thesis with the University Library.

Code, data and every script behind each number: github.com/brn-mwai/monarch. The three journal papers from this work will follow separately.